



## Short communication

Exploring the genetic mechanisms underlying amoxicillin-clavulanate resistance in waterborne *Escherichia coli*Nambram Somendro Singh<sup>a,b</sup>, Neelja Singhal<sup>b,\*</sup>, Manish Kumar<sup>b</sup>, Jugsharan Singh Virdi<sup>a</sup><sup>a</sup> Department of Microbiology, University of Delhi South Campus, New Delhi, India<sup>b</sup> Department of Biophysics, University of Delhi South Campus, New Delhi, India

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## ABSTRACT

*Escherichia coli* is a human commensal and faecal indicator bacteria which is also the etiologic agent of several nosocomial- and community-acquired infections. Amoxicillin-clavulanate (AMC) is a widely prescribed β-lactam/β-lactamase inhibitor which is used against *E. coli* infections. Resistance to AMC in *E. coli* has been primarily attributed to point mutations in *bla*<sub>TEM-1</sub> resulting in inhibitor-resistant TEM (IRT) β-lactamases. In this study, we have explored the reasons underlying AMC-resistance in waterborne *E. coli*. Most of the studies regarding IRT-producing *E. coli* have been conducted on clinical samples and studies exploring genetic mechanisms underlying AMC-resistance in aquatic *E. coli* are scarce. Since, *bla*<sub>TEM-1</sub> and several antimicrobial resistance determinants are located on mobile genetic elements they can easily disseminate among other microbes inhabiting urban waterbodies. Thus, it is important to understand the underlying mechanisms to check the dissemination of AMC-resistance in other waterborne pathogens. Our results indicated that AMC-resistant *E. coli* were susceptible to other β-lactam/β-lactamase inhibitors like, ampicillin/sulbactam and piperacillin/tazobactam. Though, *bla*<sub>TEM-1</sub> was present, none of the strains harbored point mutations which could qualify as IRT and only one strain harbored both *bla*<sub>TEM-1</sub> and *bla*<sub>OXA-1</sub>. Hyperproduction of *bla*<sub>TEM-1</sub>, presence of plasmid-mediated *ampC* or promoter/attenuator mutations in the chromosomal *ampC* might not be related to IRT-like phenotype or AMC-resistance. This suggests that other mechanisms like, increased plasmid copy numbers or gene amplification or deficiency in the expression/function of porins might be responsible for AMC-resistance in waterborne *E. coli*.

## 1. Introduction

*Escherichia coli* is a human commensal and faecal indicator bacteria which is also the etiologic agent of several nosocomial- and community-acquired infections (Briñas et al., 2002). Beta-lactams are the most efficient and widely used antibiotics against enteric bacteria, especially *E. coli* (Bush and Bradford, 2016). However, bacteria have evolved several variants of β-lactamases which hydrolyze the β-lactam antibiotics making them inefficient. To combat β-lactam resistance, β-lactamase inhibitors along with newer β-lactams were discovered. Several β-lactam/β-lactamase inhibitor combinations like, amoxicillin-clavulanate, ampicillin-sulbactam and piperacillin-tazobactam restored the efficacy of penicillins. However, extensive use of β-lactamase inhibitors led to the emergence of inhibitor-resistant variants of TEM and SHV β-lactamases which have compromised the efficacy of β-lactam/β-lactamase inhibitors.

Amoxicillin-clavulanate (AMC) is a widely used β-lactam/

β-lactamase inhibitor for treating *E. coli* infections (Mimica Matanovic et al., 2010). Resistance to AMC in *E. coli* can be due to several reasons like, hyperproduction of *bla*<sub>TEM-1</sub> due to strong promoters, presence of OXA-type β-lactamases, hyperproduction of the chromosomal AmpC cephalosporinase (Stapleton et al., 1995), deficiency in the OmpF and/or OmpC porins and, presence of inhibitor-resistant TEM (IRT) β-lactamases (Cantón et al., 2008). The IRT evolved due to point mutations in *bla*<sub>TEM-1</sub> resulting in resistance to AMC. According to a previous study, it is the most common mechanism underlying β-lactam/β-lactamase inhibitor resistance in *E. coli* (Robin et al., 2011) and, at least 19 distinct IRT have been reported (<https://card.mcmaster.ca/ontology/36023>).

The *bla*<sub>TEM-1</sub> is usually plasmid borne, but it can easily spread and disseminate among bacteria via transposition and can also insert into the chromosome (Heffron et al., 1975; Lescat et al., 2009; Bidet et al., 2016). Most of the studies regarding IRT-producing *E. coli* have been conducted on clinical samples, especially urine samples (Ding et al., 2013; Shubert et al., 2014). Though, β-lactam/β-lactamase inhibitor resistance has

\* Corresponding author.

E-mail address: [neelja30@gmail.com](mailto:neelja30@gmail.com) (N. Singhal).<https://doi.org/10.1016/j.meegid.2021.104767>

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been reported in *E. coli* isolated from waterbodies (Adesoji and Ogunjobi, 2016; Haberecht et al., 2019) the underlying reasons for  $\beta$ -lactam/ $\beta$ -lactamase inhibitor resistance have not been investigated in *E. coli* isolated from urban waterbodies, especially India. Urban waterbodies receive a diverse pool of microbial population along with waste waters originating from hospitals, industries, veterinary and agricultural fields, etc. Since, several antimicrobial resistance determinants including *bla*<sub>TEM-1</sub> are located on mobile genetic elements they can easily disseminate among the microbes inhabiting urban waterbodies. Thus, it is important to understand the antibiotic resistance and underlying mechanisms in *E. coli* inhabiting urban waterbodies to check the spread of resistance determinants among waterborne pathogens. In an earlier study we had isolated AMC-resistant strains of *E. coli* from a major urban river of India (Bajaj et al., 2015). In this study, we have tried to discern the reasons underlying AMC-resistance in these strains. Also, the susceptibility of these strains for other  $\beta$ -lactam/ $\beta$ -lactamase inhibitors like ampicillin-sulbactam and piperacillin-tazobactam was determined and, the reasons underlying  $\beta$ -lactam/ $\beta$ -lactamase inhibitor resistance were investigated.

## 2. Material and methods

### 2.1. Bacterial strains

In this study, 18 AMC-resistant strains of *E. coli* were included (Bajaj et al., 2015). These strains were isolated from the water samples collected from the stretch of river Yamuna that flows through the National Capital Region of India. The samples were collected in sterile screw-capped bottles and enriched for isolation of *E. coli* following a published protocol (Ram et al., 2007). Briefly, 100 ml of the water sample was passed through a 0.45  $\mu$ m membrane filter (Millipore, MA, USA) which was later cut into four pieces and each piece was kept for overnight incubation in 50 ml of MacConkey broth at 37 °C. A loopful of the broth culture was inoculated on the surface of MacConkey agar plates which were further incubated. Typical colonies of *E. coli* were selected and presumptively identified as *E. coli* using the API 20E strips - a standardized kit of biochemical tests for identification of enteric bacteria, (bioMérieux, France). The isolates were confirmed by 16S rDNA sequencing and serotyped at the National Salmonella and Escherichia Centre, Central Research Institute, Kasauli, Himachal Pradesh, India (Bajaj et al., 2015).

The minimal inhibitory concentration (MIC) of all the strains for AMC was determined using E-test strips for AMC (bioMérieux Inc., MO, USA) following the manufacturer's instructions. The strains were designated as susceptible/intermediate/resistant based on the MIC standard criteria specified in the guidelines of Clinical and Laboratory Standards Institute (CLSI, 2012). Of these, 18 *E. coli* strains which were resistant for AMC were investigated in this study.

### 2.2. Determination of minimal inhibitory concentrations for $\beta$ -lactam/ $\beta$ -lactamase inhibitors

The minimal inhibitory concentrations (MICs) of AMC-resistant strains for ampicillin-sulbactam and piperacillin-tazobactam were determined using E-test strips (bioMérieux Inc., MO, USA) following the manufacturer's instructions. The MIC for ampicillin-sulbactam and piperacillin-tazobactam was interpreted according to the guidelines of Clinical and Laboratory Standards Institute (CLSI, 2012).

### 2.3. Detection of inhibitor-resistant TEM (IRT) and *bla*<sub>OXA</sub> genes

The complete coding sequences (CCDS) of *bla*<sub>TEM</sub> and *bla*<sub>OXA</sub> genes were determined in the 18 AMC-resistant *E. coli* strains by PCR-amplification of the relevant genes. Published primers and PCR conditions were used for amplification of the CCDS of *bla*<sub>TEM</sub> and *bla*<sub>OXA-1</sub> (Singh et al., 2018). The sequences of the primers for PCR-amplification

of *bla*<sub>TEM</sub> were - 5'TAATAATGGTTTCTTAGACG3' (forward primer) and 5'CATGCATCTGTATAAGGGGT3' (reverse primer) and, for *bla*<sub>OXA-1</sub> were 5'GGGCGAACCCGGAGC3' (forward primer) and 5'TTTGAAATAAGGGGCGCCGA3' (reverse primer). The PCR reaction mixture (25  $\mu$ l) was prepared by mixing 2.5  $\mu$ l of 1 $\times$  buffer, 200  $\mu$ M of each of the four dNTPs, 6  $\mu$ l of template DNA, 20 pmol of forward and reverse primers and 1 U of Taq DNA polymerase. PCR cycles included an initial denaturation step of 95 °C for 5 min, 25 cycles of denaturation at 95 °C for 1 min, annealing (44 °C for *bla*<sub>TEM-1</sub> and 60 °C for *bla*<sub>OXA-1</sub>) for 30 s and extension at 72 °C for 1.5 min, followed by a final extension at 72 °C for 10 min. The PCR amplicons were electrophoresed on 1% agarose gels, stained with ethidium bromide and visualized under a UV transilluminator. The PCR amplicons were purified using Hi-Yield™ extraction kit (RBC Bioscience, New Taipei City, Taiwan) following manufacturer's instructions and sequenced using Sanger's method at a commercial facility (Invitrogen BioServices India Pvt. Ltd., Bangalore, India). Nucleotide similarity search was performed using BLASTn (NCBI).

## 3. Result and discussion

### 3.1. Phenotypic susceptibilities to $\beta$ -lactam/ $\beta$ -lactamase inhibitors

The MIC values of the *E. coli* strains indicated that all the strains were susceptible to ampicillin-sulbactam and piperacillin-tazobactam (Table 1). However, all these strains exhibited an MIC >256 mg/l for AMC (Bajaj et al., 2015), indicating that these strains were highly resistant for AMC. AMC has proved as a highly effective antibiotic against infections caused by Gram-positive, -negative and anaerobic bacteria (Bush and Johnson, 2000). Compared to other  $\beta$ -lactam/ $\beta$ -lactamase inhibitors, the frequent prescription of AMC for treating bacterial infections might be plausible reason for a higher resistance (Garg et al., 2013).

### 3.2. Sequence-based analysis of inhibitor resistant TEM (IRT) and *bla*<sub>OXA-1</sub> genes

PCR-amplification of *bla*<sub>TEM</sub> resulted in the desired amplicon of 1175 bp in all the strains. Gene sequencing of the PCR amplicons and similarity search revealed that *bla*<sub>TEM</sub> of all the strains were identical and shared 100% identity with the *bla*<sub>TEM-1</sub> sequence available in the NCBI. The *bla*<sub>TEM-1</sub> CCDS of a representative strain (KK30) was submitted in

**Table 1**  
Minimal inhibitory concentration of AMC-resistant *E. coli* strains for  $\beta$ -lactam/ $\beta$ -lactamase inhibitors.

| <i>E. coli</i> strain designation | MIC of AMC -resistant <i>E. coli</i> strains for $\beta$ -lactam/ $\beta$ -lactamase inhibitors (0.016–256 mg/l) |                         |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------|
|                                   | Ampicillin/sulbactam                                                                                             | Piperacillin/tazobactam |
| NG9                               | 4 (S)                                                                                                            | 1(S)                    |
| KP21                              | 12(S)                                                                                                            | 2(S)                    |
| NG3                               | 6(S)                                                                                                             | 1.5(S)                  |
| KK30                              | 8(S)                                                                                                             | 2(S)                    |
| WB23                              | 6(S)                                                                                                             | 1(S)                    |
| IS5                               | 8(S)                                                                                                             | 1.5(S)                  |
| NG29                              | 4(S)                                                                                                             | 2(S)                    |
| IP5N                              | 16(S)                                                                                                            | 2(S)                    |
| NG32                              | 6(S)                                                                                                             | 1.5(S)                  |
| ISI                               | 48(S)                                                                                                            | 1(S)                    |
| KKA                               | 16(S)                                                                                                            | 2(S)                    |
| PA12                              | 12(S)                                                                                                            | 3(S)                    |
| KK16                              | 32(S)                                                                                                            | 3(S)                    |
| KK31                              | 32(S)                                                                                                            | 2(S)                    |
| WB6                               | 24(S)                                                                                                            | 2(S)                    |
| IPE                               | 32(S)                                                                                                            | 8(S)                    |
| ISE                               | 32(S)                                                                                                            | 4(S)                    |
| KK45                              | 32(S)                                                                                                            | 4(S)                    |

S-sensitive.

GenBank [accession number MW183920]. None of the AMC-resistant strains harbored point mutations in *bla*<sub>TEM-1</sub> which could qualify as IRT (Drawz and Bonomo, 2010).

Among the 18 AMC-resistant strains, a PCR amplicon of 1000 bp corresponding to *bla*<sub>OXA</sub> was observed in only one strain (KK45). Sequencing and similarity search revealed that the CCDS of *bla*<sub>OXA</sub> shared a 100% sequence identity with the *bla*<sub>OXA-1</sub> gene sequence available at NCBI. The *bla*<sub>OXA-1</sub> CCDS of *E. coli* strain KK45 was submitted in GenBank [accession number MW183921].

The absence of IRTs in AMC-resistant *E. coli* strains reflects that IRTs were not prevalent in waterborne *E. coli*. Thus, our results are in concordance with the previous studies which also reported that IRTs were less prevalent in *E. coli* strains that exhibited IRT-like phenotype (Miro et al., 2002; Pérez-Moreno et al., 2004; Kiiru et al., 2012; Ortega et al., 2012). Though, an earlier study reported that majority of the *E. coli* strains which exhibited an IRT-like phenotype harbored both *bla*<sub>TEM-1</sub> and *bla*<sub>OXA-1</sub> (Kiiru et al., 2012) but in this study, only one strain (KK45) was found to harbor both *bla*<sub>TEM-1</sub> and *bla*<sub>OXA-1</sub>.

Hyperproduction of *bla*<sub>TEM-1</sub> due to strong promoters is reportedly another mechanism underlying AMC-resistance in *E. coli* (Wu et al., 1995; Pérez-Moreno et al., 2004). However, an earlier study conducted by us indicated that *bla*<sub>TEM-1</sub> in all these strains was regulated by P3 type promoters, which are considered as weak promoters (Singh et al., 2018). Though weak, P3 type promoters are the most prevalent promoters of *bla*<sub>TEM-1</sub> which were reportedly present in more than 50% of *bla*<sub>TEM-1</sub> positive, clinical AMC-resistant *E. coli* strains (Ortega et al., 2012). Interestingly, of the four different types of *bla*<sub>TEM</sub> promoters, viz. Pa/Pb, P3, P4 and P5; P3 type promoters are reportedly associated with lower expression levels of *bla*<sub>TEM-1</sub> than Pa/Pb, P4, and P5 promoters (Lartigue et al., 2002; Tristram and Nichols, 2006; Waltner-Toews et al., 2011; Sun et al., 2013). Thus, gene promoters are not likely associated with AMC-resistance in these *E. coli* strains.

Presence of plasmid-mediated *ampC* or overexpression of the constitutively expressed chromosomal *ampC* has also been reportedly associated with AMC-resistance in *E. coli* (Pérez-Moreno et al., 2004). However, results from an earlier study conducted in our laboratory indicated that of these 18 AMC-resistant *E. coli* strains, only two strains IPE and ISE harbored the plasmid mediated *ampC* gene - *bla*<sub>CMY-42</sub> (Bajaj et al., 2015). Similarly, analysis of the promoter/attenuator of the chromosomal *ampC* revealed that except in two strains, mutations were absent in the promoter/attenuator regions of *ampC* (Bajaj et al., 2015). Interestingly, the nucleotide positions at which *ampC* promoter/attenuator mutations were observed in the *E. coli* strains IPE and ISE, were unrelated to the significant overproduction of chromosomal *ampC* gene (Peter-Getzlaff et al., 2011; Bajaj et al., 2015).

In an earlier study from our laboratory we had performed conjugation assays in these AMC-resistant strains (Singh et al., 2018). Analysis of the plasmid DNA of the transconjugants revealed that *bla*<sub>TEM-1</sub> was plasmid-mediated which reflects its potential to transmit to other waterborne pathogens by horizontal gene transfer. This accentuates the importance of a precise understanding of the mechanisms underlying AMC-resistance in *E. coli* to for devising suitable intervention strategies for widespread dissemination of AMC-resistance among waterborne pathogens.

To summarize, our results suggest that though the waterborne *E. coli* were resistant to AMC, they were susceptible to other  $\beta$ -lactam/ $\beta$ -lactamase inhibitors like, ampicillin-sulbactam and piperacillin-tazobactam. Though, *bla*<sub>TEM-1</sub> was present in all the strains exhibiting IRT-like phenotype, none of the strains harbored an IRT. Hyperproduction of *bla*<sub>TEM-1</sub>, presence of plasmid-mediated *ampC* or promoter/attenuator mutations in the chromosomal *ampC* might not be related to IRT-like phenotype or AMC resistance. Thus, it would be worthwhile to explore if other mechanisms like, increased plasmid copy numbers or gene amplification i.e. increased number of *bla*<sub>TEM-1</sub> copies (Schechter et al., 2018) or deficiency in the expression/function of porins were responsible for AMC-resistance in waterborne *E. coli*. Though, our results

are preliminary, this study could serve as a useful basis for further studies to understand AMC-resistance in waterborne *E. coli*.

## Authors' contributions

Conceptualization: [Jugsharan Singh Virdi]; Methodology: [Nambram Somendro Singh]; Formal analysis and investigation: [Nambram Somendro Singh], [Neelja Singhal], [Manish Kumar]; Writing - original draft preparation: [Nambram Somendro Singh], [Neelja Singhal]; Writing - review and editing: [Neelja Singhal], [Manish Kumar]; Funding acquisition: [Jugsharan Singh Virdi]; Resources: [Jugsharan Singh Virdi]; Supervision: [Neelja Singhal], [Jugsharan Singh Virdi]. All authors contributed to the study conception and design. All authors read and approved the final manuscript.

## Declaration of Competing Interest

Authors do not have any competing interests.

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